AGRI-HEALTH AI PROJECT

Software Requirements Specification

Lumpy Skin Disease (LSD) Detection Web Application

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete, detailed, and authoritative description of the Lumpy Skin Disease (LSD) Detection Web Application. It defines all functional requirements, non-functional requirements, system architecture, data flow, user interface specifications, AI/ML model integration, deployment strategy, and acceptance criteria for the end-to-end production system.

This document is intended to serve as the single source of truth for all development, testing, and deployment activities. It shall be used by software engineers, AI/ML engineers, UI/UX designers, QA testers, DevOps engineers, and project stakeholders.

1.2 Scope

The LSD Detection Web Application is an AI-powered, browser-based diagnostic tool designed to help veterinarians, livestock officers, and farmers detect Lumpy Skin Disease in cattle using photographic images. The system encompasses:

- A React-based responsive web frontend that runs on any modern browser
- A FastAPI Python backend that serves AI model predictions via REST API
- A deep learning computer vision model (MobileNetV2 / EfficientNetV2) trained on cattle skin imagery
- Grad-CAM visual explainability layer highlighting disease regions
- Confidence scoring with actionable farmer-friendly output
- Scan history, herd management, and report generation
- Cloud deployment on scalable infrastructure (Docker + cloud hosting)
- Optional offline Progressive Web App (PWA) capability with TFLite

1.3 Document Conventions

Term / Abbreviation	Definition
LSD	Lumpy Skin Disease — a viral cattle disease caused by Capripoxvirus
SRS	Software Requirements Specification — this document
CNN	Convolutional Neural Network — deep learning architecture for image analysis
Grad-CAM	Gradient-weighted Class Activation Mapping — AI explainability technique
TFLite	TensorFlow Lite — compressed model format for mobile/edge deployment
PWA	Progressive Web App — web app with offline and native-like capabilities
API	Application Programming Interface — communication contract between services
REST	Representational State Transfer — HTTP-based API design pattern
FR	Functional Requirement
NFR	Non-Functional Requirement
UC	Use Case
UI/UX	User Interface / User Experience
CI/CD	Continuous Integration / Continuous Deployment
EDA	Exploratory Data Analysis

1.4 Intended Audience

Audience	Role	Sections of Interest
Software Engineers	Frontend & Backend Development	3, 4, 5, 6, 7
AI/ML Engineers	Model Training & Integration	3.5, 4, 7, 8
UI/UX Designers	Interface & Experience Design	5, 6
QA / Test Engineers	Testing & Validation	9, 10
DevOps Engineers	Deployment & Infrastructure	11, 12
Project Stakeholders	Business Requirements & Goals	1, 2, 3, 13
Veterinary Consultants	Domain Validation & Acceptance	2, 3.5, 9

1.5 References

- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications
- TensorFlow 2.x Documentation — https://www.tensorflow.org/api_docs
- FastAPI Official Documentation — https://fastapi.tiangolo.com
- React 18 Documentation — https://react.dev
- Mendeley Dataset: Lumpy Skin Disease Cattle Images (2023)
- OIE — World Organisation for Animal Health: LSD Disease Card
- Selvaraju, R.R. et al. (2017) Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization

2. Overall Description

2.1 Product Perspective

The LSD Detection Web Application is a standalone, cloud-hosted AI diagnostic tool that operates independently while integrating with existing veterinary workflows. It is the first product in a planned Agri-Health AI suite. The system interfaces with:

- End users via browser-based web interface (mobile and desktop)
- AI model inference engine via internal REST API calls

- Cloud storage for scan history and uploaded images
- Optional integration with veterinary management systems via webhook/API
- SMS gateway for low-connectivity result delivery (future phase)

2.2 Product Functions — High Level Summary

#	Function	Description
F-01	**Image Upload & Capture**	Accept cattle skin images via file upload, drag-and-drop, or direct camera capture on mobile devices
F-02	**Image Quality Validation**	Automatically check image quality — blur detection, size validation, brightness assessment — before inference
F-03	**AI Disease Inference**	Run deep learning model inference and classify image as Healthy, LSD Early Stage, LSD Active, or LSD Late Stage
F-04	**Confidence Scoring**	Provide per-class probability scores with an overall confidence percentage for the top prediction
F-05	**Grad-CAM Explainability**	Generate heatmap overlay on the original image showing which skin regions influenced the prediction most
F-06	**Actionable Result Display**	Translate raw model output into plain-language, colour-coded farmer-friendly recommendations
F-07	**Scan History & Herd Tracking**	Persist scan results per animal/herd with timestamps, enabling longitudinal disease monitoring
F-08	**Report Generation**	Export scan results as formatted PDF or shareable link for veterinary consultation
F-	**User Authentication**	Secure user accounts with email/password and optional OAuth (Google) sign-in

09		
F-10	**Admin Dashboard**	Monitor system usage, model performance metrics, and dataset drift indicators

2.3 User Classes and Characteristics

2.3.1 Primary Users

User Class	Technical Level	Primary Goals	Device Usage
Livestock Farmer	Low to Medium	Quick, reliable LSD detection in the field	Mobile phone, rural connectivity
Veterinarian	High	Clinical decision support, differential diagnosis	Desktop + tablet, hospital network
Livestock Officer	Medium	Surveillance, outbreak monitoring, reporting	Tablet + desktop, moderate connectivity
Agricultural Extension Worker	Low to Medium	Field screening, farmer education, referrals	Mobile, low connectivity

2.3.2 Secondary Users

- System Administrator — manages user accounts, monitors system health, configures model versions
- Researcher / Data Scientist — accesses model performance dashboards, triggers retraining pipelines
- Government Veterinary Authority — views aggregated outbreak heatmaps and compliance reports

2.4 Operating Environment

Component	Specification
Frontend Runtime	Modern web browsers: Chrome 90+, Firefox 88+, Safari 14+, Edge 90+. Mobile: Chrome for Android, Safari for iOS
Backend Runtime	Python 3.11+, FastAPI 0.104+, TensorFlow 2.13+, running on Linux (Ubuntu 22.04 LTS)

Minimum Device (Mobile)	Android 8.0+ or iOS 13+, 3GB RAM, 5MP camera, 500MB free storage
Network Requirements	Core prediction: minimum 3G (500 kbps). Offline mode: no network required after initial load
Cloud Infrastructure	Docker containers on Google Cloud Run / AWS ECS / Render.com. PostgreSQL 15 for data persistence
AI Model Environment	CUDA 12.x GPU (training only). CPU inference acceptable for <2s latency on TFLite quantized model

2.5 Design and Implementation Constraints

- The AI model must never claim to provide definitive clinical diagnosis; all outputs must be framed as decision support.
- All patient (animal) data must be stored with GDPR-compliant data handling; no personally identifiable farmer data shall be shared with third parties.
- The frontend must achieve a Lighthouse accessibility score of 90+ to support diverse literacy levels.
- Model inference latency must not exceed 5 seconds on cloud, or 3 seconds on-device with TFLite.
- The system must function in Progressive Web App (PWA) mode with a cached TFLite model for offline use.
- All source code must be maintained in version-controlled repositories with mandatory code review before merging.

2.6 Assumptions and Dependencies

- It is assumed that users have access to a functional smartphone camera capable of capturing at minimum 5MP images.
- The AI model training is assumed to be completed prior to backend deployment; model weights are treated as a versioned artefact.
- Cloud provider (GCP/AWS/Render) uptime SLA of 99.5% is assumed for availability calculations.
- The initial dataset (Mendeley, ~1024 images) is assumed to be representative of the geographic target population; model performance may degrade in regions with distinct cattle breeds.
- Future multi-language support depends on availability of translation resources for regional Indian languages.

3. System Architecture

3.1 High-Level Architecture Overview

The application follows a clean three-tier architecture with an additional AI inference service layer. The tiers are loosely coupled and communicate via well-defined REST APIs, enabling independent scaling, deployment, and testing of each layer.

SYSTEM ARCHITECTURE DIAGRAM

[CLIENT TIER]

React SPA | PWA (TFLite offline) | Mobile Browser

↕ HTTPS / REST

[APPLICATION TIER]

FastAPI Backend | Auth Service | Business Logic | Image Preprocessor

↕ Internal API Call

[AI INFERENCE LAYER]

TF Model Loader | Grad-CAM Engine | Confidence Scorer | Result Formatter

↕ SQL / ORM

[DATA TIER]

PostgreSQL DB | Cloud Storage (images/models) | Redis Cache

3.2 Technology Stack

Layer	Technology	Justification
Frontend	**React 18 + TypeScript + Tailwind CSS + Vite**	Component-based, type-safe, fast build, utility-first styling for rapid UI development
State Management	**Zustand + React Query (TanStack)**	Lightweight global state, automatic API caching and synchronisation
HTTP Client	**Axios with interceptors**	Request/response transformation, automatic auth token injection, error handling
AI (On-Device)	**TensorFlow.js + TFLite (WASM)**	Offline inference in browser; fallback when API unavailable
Backend Framework	**FastAPI (Python 3.11)**	Async-native, auto-generated OpenAPI docs, excellent performance, type annotations

AI/ML Runtime	**TensorFlow 2.13 + Keras**	Mature ecosystem, Grad-CAM support, TFLite conversion, model serving
Image Processing	**OpenCV + Pillow + NumPy**	Robust preprocessing pipeline — resize, normalise, augment, validate
Database ORM	**SQLAlchemy 2.0 + Alembic**	Type-safe queries, async support, schema migration management
Database	**PostgreSQL 15**	ACID-compliant, JSON support for flexible scan metadata, proven reliability
Caching	**Redis 7**	Session storage, prediction caching, rate-limit counters
Auth	**JWT (python-jose) + bcrypt**	Stateless authentication, industry-standard, works with PWA and API clients
Containerisation	**Docker + Docker Compose**	Reproducible environments, easy deployment, service orchestration
Cloud Deployment	**Google Cloud Run / Render**	Serverless containers, auto-scaling, HTTPS by default, pay-per-use
CI/CD	**GitHub Actions**	Automated testing, linting, Docker build, and cloud deployment on push

3.3 Directory & Module Structure

Backend Structure (FastAPI)

```
lsd_backend/
├── main.py            # App entry, CORS,
routers
├── config.py          # Settings, env
vars
├── models/
│   ├── lsd_model.py   # TF model loader
│   ├── gradcam.py     # Grad-CAM engine
│   └── predictor.py   # Inference
pipeline
├── routers/
│   ├── predict.py     # /predict endpoint
│   ├── auth.py        # /auth endpoints
│   ├── history.py     # /scans endpoints
│   └── admin.py       # /admin endpoints
├── schemas/
│   ├── prediction.py  # Pydantic models
│   └── user.py
├── db/
```

Frontend Structure (React)

```
lsd_frontend/
├── src/
│   ├── App.tsx
│   ├── pages/
│   │   ├── Home.tsx
│   │   ├── Scan.tsx       # Main scan UI
│   │   ├── Results.tsx    # Result + heatmap
│   │   ├── History.tsx    # Scan history
│   │   └── Dashboard.tsx  # Admin
│   ├── components/
│   │   ├── ImageUploader.tsx
│   │   ├── ResultCard.tsx
│   │   ├── HeatmapViewer.tsx
│   │   ├── ConfidenceBar.tsx
│   │   └── ActionPanel.tsx
│   ├── services/
│   │   ├── api.ts         # Axios calls
│   │   └── tflite.ts      # On-device model
│   ├── store/
```

```
│   ├── database.py    # SQLAlchemy
session
│   └── crud.py        # DB operations
├── utils/
│   ├── image.py       # Preprocessing
│   └── validators.py  # Quality checks
├── Dockerfile
└── requirements.txt
```

```
│   │   └── useScanStore.ts
│   └── utils/
│       └── imageUtils.ts
├── public/
│   └── lsd_model.tflite   # Bundled model
└── package.json
```

4. Functional Requirements

4.1 User Authentication & Authorisation

ID	Requirement	Description	Priority
FR-AUTH -01	**User Registration**	System shall allow new users to register with name, email, password, and role selection (Farmer / Vet / Officer). Email verification required before account activation.	HIGH
FR-AUTH -02	**User Login**	System shall authenticate users via email/password. Successful login returns a JWT access token (expiry: 30 min) and refresh token (expiry: 7 days).	HIGH
FR-AUTH -03	**OAuth Login**	System shall support Google OAuth 2.0 sign-in as an alternative authentication method.	MEDIUM
FR-AUTH -04	**Password Reset**	System shall provide a secure password reset flow via time-limited email link (15-minute expiry).	HIGH
FR-AUTH -05	**Role-Based Access**	System shall enforce RBAC: Farmers see own scans only; Vets see herd-level data; Officers see regional data; Admins have full access.	HIGH
FR-AUTH -06	**Session Management**	System shall invalidate tokens on logout and support token refresh without re-authentication during active sessions.	HIGH

4.2 Image Upload & Quality Validation

ID	Requirement	Description	Priority
FR-IMG-01	**File Upload**	System shall accept image uploads via: drag-and-drop zone, file picker dialog, and mobile camera capture. Accepted formats: JPEG, PNG, WEBP, BMP. Maximum file size: 20 MB.	HIGH
FR-IMG-02	**Blur Detection**	System shall compute Laplacian variance of the uploaded image. If variance < 100 (configurable), display a warning: 'Image appears blurry. For best results, please retake the photo.'	HIGH
FR-IMG-03	**Brightness Check**	System shall reject images where mean pixel brightness < 30 (too dark) or > 240 (overexposed), with specific user guidance for correction.	HIGH
FR-IMG-04	**Minimum Resolution**	System shall reject images smaller than 100x100 pixels and recommend minimum 500x500 pixels for best accuracy.	MEDIUM
FR-IMG-05	**Preview & Crop**	System shall display a live preview of the uploaded image and allow optional cropping to focus on the affected skin area before submission.	MEDIUM
FR-IMG-06	**Multi-Image Scan**	System shall accept up to 5 images per scan session for the same animal, running inference on each and presenting an aggregated result.	LOW

4.3 AI Model Inference & Classification

ID	Requirement	Description	Priority
FR-AI-01	**Classification Output**	Model shall classify each image into one of four classes: (1) Healthy, (2) LSD Early Stage, (3) LSD Active Stage, (4) LSD Late/Healing Stage.	HIGH
FR-AI-02	**Confidence Score**	System shall return per-class probability scores (0.00–1.00) and display the top prediction with its confidence percentage to the user.	HIGH

FR-AI-03	Grad-CAM Heatmap	System shall generate a Grad-CAM heatmap overlay on the original image for all predictions. Heatmap shall use a green-yellow-red colour scale indicating regions of interest.	HIGH
FR-AI-04	Uncertainty Threshold	If the top prediction confidence is below 70%, system shall display: 'Low Confidence Result — please retake photo or consult a veterinarian.' The result shall still be shown.	HIGH
FR-AI-05	Inference Mode Selection	System shall automatically select cloud inference (API) when online and on-device TFLite inference when offline, with visible mode indicator in UI.	HIGH
FR-AI-06	Model Versioning	API shall expose the current model version in every prediction response. Admin dashboard shall display model version, training date, and validation accuracy.	MEDIUM
FR-AI-07	Batch Inference	API shall support batch inference for up to 10 images per request, returning ordered results array.	LOW

4.4 Result Display & Actionable Guidance

ID	Requirement	Description	Priority
FR-RES-01	Colour-coded Severity	Results shall use a consistent colour system: Green (Healthy), Yellow (Early/Monitor), Orange (Active/Urgent), Red (Severe/Emergency) with corresponding icons.	HIGH
FR-RES-02	Plain Language Output	System shall translate model output into farmer-friendly language. Example: 'LSD Active Stage Detected (94% confidence) — Isolate this animal immediately and contact a veterinarian.'	HIGH
FR-RES-03	Action Checklist	For positive LSD detections, system shall display a prioritised checklist: (1) Isolate animal, (2) Contact vet, (3) Notify local authority, (4) Vaccinate herd.	HIGH
FR-RES-04	Veterinary Contact	Result page shall include a prominent 'Find Nearest Vet' button that opens a map view or contact directory based on user location.	MEDIUM

FR-RES-05	**Side-by-Side View**	Result page shall display original image and Grad-CAM heatmap side by side, with a toggle slider to blend between them.	MEDIUM
FR-RES-06	**Share / Export**	User shall be able to share result via link, download as PDF report, or send via WhatsApp/SMS directly from the result page.	MEDIUM

4.5 Scan History & Herd Management

ID	Requirement	Description	Priority
FR-HIS-01	**Scan Persistence**	Every completed scan shall be automatically saved to the user's account with: timestamp, image thumbnail, prediction result, confidence score, animal tag (optional).	HIGH
FR-HIS-02	**Herd Registration**	Users shall be able to create a herd profile with: farm name, location (GPS/manual), herd size, cattle breed, and assign scans to specific animals via tag/ID.	MEDIUM
FR-HIS-03	**Disease Timeline**	For each registered animal, system shall display a timeline chart showing scan history and disease progression or recovery over time.	MEDIUM
FR-HIS-04	**Search & Filter**	History view shall support filtering by date range, result type (Healthy/LSD), confidence level, and animal ID. Full-text search over scan notes.	MEDIUM
FR-HIS-05	**Data Export**	User shall be able to export full scan history as CSV or PDF for veterinary records or insurance claims.	MEDIUM

5. API Specification

5.1 Base URL & Versioning

Base URL	https://api.lsddetect.com/v1

All endpoints are versioned under /v1/. The API follows REST conventions. All request and response bodies use JSON (except multipart/form-data for image uploads). All endpoints require a valid JWT Bearer token in the Authorization header unless marked as public.

5.2 Authentication Endpoints

Method	Endpoint	Description	Request Body	Response
POST	/auth/register	Register a new user account. Sends verification email.	`{ name, email, password, role }`	`201: { user_id, message }`
POST	/auth/login	Authenticate and receive JWT tokens. [PUBLIC]	`{ email, password }`	`200: { access_token, refresh_token, user }`
POST	/auth/refresh	Exchange refresh token for new access token.	`{ refresh_token }`	`200: { access_token }`
POST	/auth/logout	Invalidate the current session tokens.	`{ refresh_token }`	`200: { message }`
POST	/auth/reset-password	Send password reset email. [PUBLIC]	`{ email }`	`200: { message }`

5.3 Prediction Endpoints

Method	Endpoint	Description	Request	Response
POST	/predict/single	Run LSD inference on a single image.	multipart/form-data: file, animal_id (opt)	200: PredictionResponse

		Returns class, confidence, Grad-CAM heatmap (base64), and recommendations.		
POST	**/predict/batch**	Run inference on 2–10 images simultaneously. Returns array of PredictionResponse objects.	multipart/form-data: files[] (max 10)	200: PredictionResponse[]
GET	**/predict/models**	List available model versions and their validation metrics.	—	200: ModelInfo[]

5.3.1 PredictionResponse Schema

```
{
  "scan_id":          "uuid-v4",
  "timestamp":        "2026-02-12T10:30:00Z",
  "model_version":    "mobilenetv2-v2.1.0",
  "predicted_class": "LSD Active Stage",
  "class_index":      2,
  "confidence":       0.9412,
  "all_probabilities": {
    "Healthy":           0.0320,
    "LSD Early Stage":   0.0268,
    "LSD Active Stage": 0.9412,
    "LSD Late Stage":    0.0000
  },
  "gradcam_heatmap":  "...",
  "severity_level":   "HIGH",
  "recommendations":  ["Isolate animal immediately","Contact veterinarian within 24
hours","Report to livestock authority"],
  "low_confidence_warning": false,
  "image_quality":    { "blur_score": 412.3, "brightness": 128.7, "passed": true }
}
```

5.4 Scan History Endpoints

Method	Endpoint	Description	Query Params / Body
GET	**/scans**	List all scans for authenticated user with pagination.	page, limit, filter_class, date_from, date_to

GET	/scans/{scan_id}	Retrieve full details of a specific scan including heatmap.	—
DELETE	/scans/{scan_id}	Permanently delete a scan record and associated image.	—
GET	/scans/animal/{animal_id}	Get all scan history for a specific tagged animal.	page, limit
GET	/scans/export	Export scan history as CSV or PDF.	format (csv|pdf), date_from, date_to

6. UI/UX Specifications

6.1 Design Principles

Principle	Implementation Detail
Accessibility-First	WCAG 2.1 AA compliance. Minimum 4.5:1 colour contrast ratio. All interactive elements keyboard-navigable. ARIA labels on all form inputs and buttons.
Low-Literacy Friendly	All result states use icons alongside text. Critical actions (Isolate animal, Call vet) use large, unmistakable buttons with icon + label. Optional voice readout of results.
One-Tap Core Flow	From home screen to receiving a prediction result must be achievable in 3 taps or fewer on mobile.
Offline-Aware	Persistent banner when in offline mode. All previously loaded scans accessible without internet. Clear visual indicator of inference mode (cloud vs on-device).
Trust & Transparency	Confidence score always visible. Grad-CAM heatmap always available. Disclaimer 'AI-assisted, not a clinical diagnosis' on every result card.
Responsive Design	Mobile-first design. Breakpoints: 320px (min), 768px (tablet), 1024px (desktop). No horizontal scroll on any viewport.

6.2 Colour System

Semantic Role	Hex Code	Swatch	Usage
Primary Brand	#1B3A6B		Navigation bar, primary headings, logo text
Action Blue	#2563EB		Primary CTA buttons, links, active states
Success / Healthy	#065F46		Healthy result card background, pass indicators
Warning / Early LSD	#92400E		Early stage warning banners, medium alerts
Danger / Active LSD	#991B1B		Active/Severe LSD result cards, error states
Background	#F8FAFC		Page background, card surfaces
Text Primary	#0F172A		Body text, headings
Text Secondary	#64748B		Captions, labels, helper text

6.3 Core Screen Specifications

6.3.1 Scan Screen (Primary)

The Scan screen is the application entry point and highest-traffic screen. It must be designed for speed, clarity, and friction-free image capture.

- Header: App logo + hamburger menu. Current inference mode badge (Cloud/Offline).
- Upload Zone: Full-width dashed border zone. Icon + text 'Tap to photograph or upload cattle skin'. Accepts drag-and-drop on desktop.
- Camera Quick-Capture: Prominent 'Use Camera' button for mobile users. Opens device camera directly.
- Image Preview: Thumbnail with quality score overlay once image is selected. 'Retake' and 'Proceed' buttons.
- Quality Warning Banner: Non-blocking yellow banner if blur/brightness issues detected, with specific improvement tip.
- Analyse Button: Full-width green button 'Analyse for LSD'. Disabled until valid image is loaded.
- Loading State: Animated cow icon with progress stages: 'Uploading...', 'Analysing...', 'Generating heatmap...'

6.3.2 Results Screen

The Results screen must communicate the prediction outcome immediately and clearly, without requiring the user to read fine print.

- Result Banner: Full-width colour-coded banner at top with disease classification and confidence badge.
- Image Comparison: Side-by-side original vs heatmap, with a blend slider (0% original to 100% heatmap).
- Confidence Breakdown: Visual bar chart showing probability per class. Animated on load.
- Action Panel: Conditionally shown action checklist for LSD detections. Each item checkable by user.
- Disclaimer: Subtle bottom banner 'This is AI-assisted screening, not a clinical diagnosis.'
- Secondary Actions: Share, Save to History, Find Vet, Download Report buttons.

6.3.3 History Screen

- Grid or list view of past scans, sorted by date descending.
- Each scan card: Thumbnail, result badge, date, animal tag, confidence.
- Filter bar: Date range picker, class filter chips (All / Healthy / LSD), search field.
- Timeline view per animal when filtered by animal ID, showing disease progression graph.

7. AI/ML Model Specification

7.1 Dataset Specification

Parameter	Value	Notes
Primary Dataset	Mendeley LSD Dataset	1,024 images — 700 Normal Skin, 324 Lumpy Skin
Dataset Version	w36hpf86j2 (Version 1)	DOI verified, vetted by domain experts
Image Format	JPEG / PNG	Standardised to JPEG RGB during preprocessing
Image Resolution	Varies (256×256 min)	Resized to 224×224 for MobileNetV2, 384×384 for EfficientNetV2
Class Distribution	Healthy: 68.4% | LSD: 31.6%	Imbalance ratio 2.16:1 — addressed via class weights

Training Split	70% Train / 15% Val / 15% Test	Stratified split to maintain class ratio
Augmentation	Rotation ±20°, Flip H, Zoom 20%, Brightness ±20%, Shear 15%	Applied only to training set; 5× effective dataset size
Data Sources (Future)	Field survey images, veterinary hospitals	Target: 10,000+ images across 5 cattle breeds

7.2 Model Architecture Selection

Architecture	Accuracy	Size (MB)	Inference (CPU)	Deployment
MobileNetV2 ✅ Baseline	91-93%	14 MB	~0.9s (TFLite)	Mobile + Web + API
EfficientNetB0 ✅ Recommended	93-95%	29 MB	~1.4s (TFLite)	Web + API
EfficientNetV2-S 🏆 Best	95-97%	84 MB	~2.1s (API only)	Cloud API only
ResNet50V2	90-92%	98 MB	~2.6s	Cloud API only
ViT-Small	94-96%	87 MB	~3.1s	Cloud API only

7.3 Training Configuration

Parameter	Phase 1 (Head Only)	Phase 2 (Fine-Tune)
Base Model Frozen	Yes (100% frozen)	No (top 30 layers unfrozen)
Optimizer	Adam	Adam
Learning Rate	1e-3	1e-5 (10x smaller)
Loss Function	Categorical Cross-Entropy (label_smoothing=0.1)	Same
Epochs (max)	15	25
Early Stopping Patience	5 epochs	7 epochs
Batch Size	32	32

Class Weights	Applied (Lumpy: 2.16x)	Applied
Scheduler	ReduceLROnPlateau (factor=0.5, patience=3)	Same

7.4 Model Performance Requirements

Metric	Minimum Threshold	Target	Current (Baseline)
Overall Accuracy	**88%**	**≥ 95%**	~91-93%
Sensitivity (LSD Recall)	**93%**	**≥ 97%**	~89-94%
Specificity (Healthy Recall)	**87%**	**≥ 92%**	~93-96%
F1-Score (Weighted)	**88%**	**≥ 94%**	~91%
AUC-ROC	**0.93**	**≥ 0.97**	~0.94
False Negative Rate (LSD)	**≤ 7%**	**≤ 3%**	~6-11%
False Positive Rate (Healthy)	**≤ 13%**	**≤ 8%**	~4-7%
Inference Latency (Cloud)	**≤ 5s**	**≤ 2s**	~1.5s (MobileNet)
Inference Latency (On-Device)	**≤ 5s**	**≤ 3s**	~0.9s (TFLite)

8. Non-Functional Requirements

8.1 Performance Requirements

ID	Requirement	Metric / Measurement	Priority
NFR-PERF-01	Page Load Time — The application initial load (LCP) shall complete within 2.5 seconds on a 4G connection.	Lighthouse Core Web Vitals: LCP ≤ 2.5s	**HIGH**

NFR-PERF-02	API Response Time — The /predict/single endpoint shall return a complete response (including Grad-CAM) within 5 seconds for 95th percentile of requests.	p95 API latency ≤ 5000ms measured at load balancer	**HIGH**
NFR-PERF-03	Concurrent Users — The system shall support 500 simultaneous active users without performance degradation.	Load test: 500 virtual users, error rate < 1%, p95 latency < 8s	**HIGH**
NFR-PERF-04	Database Query Time — All database read queries shall complete within 200ms; write queries within 500ms.	Query execution plan analysis + APM monitoring	**MEDIUM**

8.2 Reliability & Availability

- NFR-REL-01: System uptime shall be ≥ 99.5% per calendar month (measured over rolling 30-day window), excluding pre-announced maintenance windows.
- NFR-REL-02: The system shall implement automatic retry logic for failed API calls (max 3 retries with exponential backoff: 1s, 2s, 4s).
- NFR-REL-03: Database backups shall be performed daily (automated), with point-in-time recovery capability for the last 7 days.
- NFR-REL-04: The system shall have a fully documented disaster recovery procedure with a Recovery Time Objective (RTO) of ≤ 4 hours and Recovery Point Objective (RPO) of ≤ 24 hours.

8.3 Security Requirements

ID	Requirement	Implementation
NFR-SEC-01	**Transport Security**	All HTTP traffic shall be redirected to HTTPS. TLS 1.2+ enforced. HSTS header with max-age=31536000.
NFR-SEC-02	**Password Security**	Passwords hashed with bcrypt (cost factor 12). Minimum 8 characters, at least 1 uppercase, 1 digit.
NFR-SEC-03	**API Rate Limiting**	Unauthenticated endpoints: 20 req/min. Authenticated: 100 req/min. Predict endpoint: 30 req/min. Exceeding returns HTTP 429.

NFR-SEC-04	**Input Validation**	All file uploads validated for MIME type and file header magic bytes (not just extension). Images scanned for embedded malware signatures.
NFR-SEC-05	**Data Privacy**	Uploaded images are stored encrypted at rest (AES-256). User data never shared with third parties. GDPR-compliant deletion on request.
NFR-SEC-06	**Vulnerability Scanning**	Automated OWASP dependency scanning on every CI/CD push. Critical/High CVEs block deployment until resolved.
NFR-SEC-07	**Logging & Auditing**	All authentication events, prediction requests, and admin actions logged with timestamp, user ID, IP address. Logs retained 90 days.

8.4 Scalability

- The backend shall be deployed as stateless Docker containers, enabling horizontal scaling via container orchestration (Cloud Run auto-scaling or Kubernetes HPA).
- The AI model shall be loadable from cloud storage on container startup, enabling model updates without redeploying application code.
- The database connection pool shall be configurable (default: min 5, max 20 connections) and database connection strings shall be injected via environment variables.
- CDN (Cloudflare or similar) shall serve all static frontend assets, reducing origin server load and improving global latency.

9. Data Models & Database Schema

9.1 Core Entities

9.1.1 users Table

Column	Type	Constraints	Description
id	`UUID`	PK, NOT NULL	Auto-generated UUID primary key
email	`VARCHAR(255)`	UNIQUE, NOT NULL	User's email address — used for login

password_hash	VARCHAR(255)	NOT NULL	bcrypt hashed password (cost 12)
full_name	VARCHAR(255)	NOT NULL	User's display name
role	ENUM	NOT NULL	farmer | vet | officer | admin
is_verified	BOOLEAN	DEFAULT FALSE	Email verification status
created_at	TIMESTAMPTZ	NOT NULL	Account creation timestamp (UTC)
updated_at	TIMESTAMPTZ	NOT NULL	Last update timestamp (UTC)
last_login	TIMESTAMPTZ	NULLABLE	Most recent successful login

9.1.2 scans Table

Column	Type	Constraints	Description
id	UUID	PK, NOT NULL	Auto-generated UUID primary key
user_id	UUID	FK → users.id	Owner of this scan
animal_id	VARCHAR(100)	NULLABLE	Optional farmer-assigned animal tag/ID
herd_id	UUID	FK → herds.id, NULLABLE	Associated herd (if registered)
image_path	VARCHAR(512)	NOT NULL	Cloud storage path to original image
predicted_class	VARCHAR(100)	NOT NULL	Top predicted class label
confidence	FLOAT(4)	NOT NULL, 0.0–1.0	Prediction confidence score
all_probabilities	JSONB	NOT NULL	JSON of all class probabilities
gradcam_path	VARCHAR(512)	NULLABLE	Cloud storage path to heatmap image
model_version	VARCHAR(50)	NOT NULL	Model version used for inference
severity_level	ENUM	NOT NULL	HEALTHY | LOW | MEDIUM | HIGH | CRITICAL
image_quality	JSONB	NULLABLE	Blur score, brightness, quality flags
notes	TEXT	NULLABLE	User-added notes for this scan

created_at	`TIMESTAMPTZ`	NOT NULL	Scan timestamp (UTC)

10. Testing Requirements

10.1 Testing Strategy Overview

Test Level	Scope	Tools	Coverage Target
Unit Tests	Individual functions, utilities, model helpers	pytest (backend), Vitest (frontend)	≥ 85% line coverage
Integration Tests	API endpoints, DB operations, ML pipeline	pytest + httpx, TestContainers	All critical paths
End-to-End Tests	Full user flows: upload → predict → result	Playwright (browser automation)	10 core user journeys
Model Evaluation Tests	Accuracy, sensitivity, specificity on holdout set	Custom evaluation notebook	Must pass thresholds in §7.4
Load / Performance Tests	API under concurrent load, memory usage	Locust, k6	500 VUs, p95 < 8s
Security Tests	SQL injection, XSS, auth bypass, file upload	OWASP ZAP, bandit (Python)	No Critical/High findings
Accessibility Tests	WCAG 2.1 AA compliance	axe-core, Lighthouse	Score ≥ 90
UAT (User Acceptance)	Field testing with 50 real farmers/vets	Structured test scripts + surveys	≥ 80% user satisfaction

10.2 Critical Test Cases

TC-ID	Test Case	Expected Result	Priority
TC-01	Upload a high-quality image of LSD-infected cattle and submit for analysis.	Prediction returns 'LSD Active Stage' with confidence ≥ 70%. Grad-CAM heatmap	**CRITICAL**

		generated. Action checklist displayed.	
TC-02	Upload a blurry image (Laplacian < 100).	Warning message displayed: 'Image too blurry'. Inference still runs. Low confidence warning shown if applicable.	HIGH
TC-03	Upload a healthy cattle image.	Prediction returns 'Healthy' with confidence ≥ 70%. Green result card. 'Keep monitoring' message shown.	CRITICAL
TC-04	Submit API request without auth token.	HTTP 401 Unauthorized returned. No prediction data exposed.	HIGH
TC-05	Exceed rate limit (>30 predict requests/min).	HTTP 429 returned with Retry-After header.	HIGH
TC-06	Use app in offline mode (no internet).	TFLite on-device inference runs. Result returned within 3s. Offline mode badge visible.	HIGH
TC-07	Upload non-image file (e.g., PDF, EXE).	HTTP 422 returned. Error message: 'Invalid file type. Please upload JPEG or PNG.'	HIGH
TC-08	Load test: 500 concurrent predict requests.	p95 latency < 8s. Error rate < 1%. No server crashes.	HIGH
TC-09	Delete a scan record and verify it is removed.	GET /scans/{deleted_id} returns HTTP 404. Image removed from cloud storage.	MEDIUM
TC-10	Export scan history as PDF.	PDF downloaded with correct data, timestamps, result summaries, and LSD images thumbnails.	MEDIUM

11. Deployment & DevOps

11.1 Deployment Architecture

- Frontend: Built as static assets (Vite build), deployed to CDN edge network (Cloudflare Pages or Netlify). PWA service worker caches app shell and TFLite model.
- Backend: Dockerized FastAPI application deployed as auto-scaling serverless container on Google Cloud Run. Min instances: 1, Max: 10.
- Database: Managed PostgreSQL instance (Cloud SQL or Supabase). Private VPC connection to backend. Daily automated backups.
- Model Storage: Versioned model files stored in Cloud Storage bucket. Loaded by backend on startup via environment variable MODEL_PATH.
- Secrets Management: All API keys, DB passwords, JWT secrets stored in Google Secret Manager / environment secrets. Never hardcoded.

11.2 CI/CD Pipeline (GitHub Actions)

Stage	Trigger	Steps
1	**Every push to any branch**	Lint (ESLint + Ruff) → Unit tests (pytest + Vitest) → Type check (mypy + tsc)
2	**Pull Request to main**	All Stage 1 + Integration tests → Security scan (bandit + npm audit) → Coverage report
3	**Merge to main**	All Stage 2 + Docker build → Push to Container Registry → Deploy to Staging environment
4	**Tag release (v*.*.*)**	All Stage 3 + E2E tests on Staging → Manual approval gate → Deploy to Production → Post-deploy health check

11.3 Environment Configuration

Variable	Staging Value	Production Value
`DATABASE_URL`	`postgresql://user:pass@staging-db/lsd_stage`	`postgresql://user:pass@prod-db/lsd_prod`
`MODEL_PATH`	`gs://lsd-models-staging/mobilenetv2-v2.1.0/`	`gs://lsd-models-prod/efficientnetv2-v3.0.0/`
`JWT_SECRET_KEY`	`<staging-secret>`	`<prod-secret> (rotated quarterly)`
`ALLOWED_ORIGINS`	`https://staging.lsddetect.com`	`https://lsddetect.com`
`REDIS_URL`	`redis://staging-redis:6379`	`redis://prod-redis-cluster:6379`
`LOG_LEVEL`	`DEBUG`	`WARNING`

RATE_LIMIT_PER_MIN	200	100

12. Ethical Considerations & Compliance

12.1 Medical/Veterinary AI Ethics

- The system shall always present its output as AI-assisted decision support, never as a definitive clinical diagnosis. This disclaimer must appear on every results screen.
- The system shall never recommend a specific pharmaceutical treatment. It shall recommend consulting a licensed veterinarian for treatment decisions.
- False negative rates must be actively monitored. If the system's false negative rate exceeds 8% in production (detected via vet feedback loop), the model shall be automatically flagged for retraining before it can be used further.
- All predictions below 70% confidence must explicitly communicate uncertainty to the user in plain language.

12.2 Data Privacy & GDPR Compliance

- Uploaded images are the property of the user. They are stored encrypted and used only to provide the prediction service. They are never used to retrain the model without explicit opt-in consent.
- Users may request full data deletion at any time via Settings > Delete My Data. Deletion must be completed within 30 days.
- Users may export all their data (scans, results, profile) in machine-readable JSON format.
- GPS/location data attached to scans is stored separately and users may opt out of location tracking.
- Minors (under 18) may not create accounts. Age verification is enforced during registration.

12.3 Bias & Fairness

- The initial dataset is predominantly from South Asian cattle breeds. The system must display a geographic scope disclaimer: 'Model validated on South Asian cattle populations. Accuracy may vary for other breeds.'
- Model performance must be monitored separately for different cattle breeds as more diverse data becomes available.
- The feedback collection mechanism must be equally accessible to users of all literacy levels (voice feedback option).

13. Acceptance Criteria & Project Milestones

13.1 Release Milestones

Milestone	Timeline	Deliverables	Acceptance Criteria
MVP v0.1	**Week 2–3**	Working Colab notebook, trained baseline model (MobileNetV2), Grad-CAM implementation	Accuracy ≥ 88%, Grad-CAM visualisation working on sample images
Alpha v0.5	**Week 4–6**	FastAPI backend with /predict endpoint, PostgreSQL schema, Docker compose setup	API returns valid PredictionResponse within 5s; all unit tests pass
Beta v0.8	**Week 7–9**	React frontend: Scan, Results, History screens; JWT auth; cloud deployment on Staging	E2E flow: upload → predict → result works end-to-end on Staging
RC v0.9	**Week 10–11**	Full feature set including export, herd management, admin dashboard; load tested	All NFRs met on Staging; UAT with 20 test users completed
v1.0 Launch	**Week 12**	Production deployment, monitoring dashboards, documentation, App Store submission (PWA)	All acceptance criteria in §13.2 met; sign-off from stakeholders

13.2 Production Release Acceptance Criteria

The system shall be deemed ready for v1.0 production release only when ALL of the following criteria are satisfied:

✓	Acceptance Criterion	Verification Method
AC-01	Model achieves ≥ 93% overall accuracy on held-out test set (never seen during training)	Automated evaluation notebook run on test set
AC-02	Model sensitivity (LSD detection rate) ≥ 95%	Confusion matrix analysis
AC-03	API /predict/single returns valid response within 5s for 95% of requests under load	Locust load test at 500 VUs for 10 minutes
AC-04	All unit, integration, and E2E tests pass with 0 failures	CI/CD pipeline green on main branch

AC-05	No Critical or High severity security vulnerabilities open	OWASP ZAP + bandit scan reports reviewed
AC-06	Lighthouse score ≥ 90 for Performance, Accessibility, Best Practices	Lighthouse CI on production URL
AC-07	Offline mode (TFLite) produces predictions within 5s without network	Manual test on Android device with flight mode
AC-08	Field UAT with ≥ 50 farmers/vets shows ≥ 80% rate the app as helpful	Survey results tabulated and reviewed
AC-09	Full GDPR data deletion request flow verified end-to-end	Manual test: register → scan → request deletion → verify removal
AC-10	All API endpoints documented in OpenAPI (Swagger) spec with examples	FastAPI /docs endpoint review

13.3 Future Roadmap (Post v1.0)

Phase	Feature	Description
v1.1	**Multi-Language**	Hindi, Marathi, Gujarati, Tamil UI + voice output using Google TTS API
v1.2	**AI Chatbot**	Integrated LLM chatbot providing post-diagnosis guidance, treatment FAQs, vet referral
v1.3	**Disease Heatmap**	Regional/national outbreak heatmap using anonymised, aggregated scan data
v1.4	**Multi-Disease**	Expand model to detect Foot-and-Mouth Disease, Mastitis, BRD using additional datasets
v2.0	**Native Mobile App**	Flutter-based Android and iOS app with enhanced camera guidance, AR lesion marking

14. Appendix

14.1 Revision History

Version	Date	Author	Description of Changes

1.0	Feb 2026	Project Team	Initial draft. Complete SRS for LSD Detection Web Application v1.0.

14.2 Glossary of AI/ML Terms

Term	Definition
Transfer Learning	Machine learning technique where a model pre-trained on a large dataset (ImageNet) is fine-tuned on a smaller domain-specific dataset (LSD cattle images).
Grad-CAM	Gradient-weighted Class Activation Mapping. Generates heatmaps showing which spatial regions of an input image most influenced a neural network's classification decision.
Sensitivity / Recall	True Positive Rate: proportion of actual LSD cases that the model correctly identifies. Formula: TP / (TP + FN). Critical for medical screening systems.
Specificity	True Negative Rate: proportion of actually healthy animals correctly identified as healthy. Formula: TN / (TN + FP).
Confidence Score	The softmax probability output for the predicted class, representing the model's estimated likelihood (0–1) that its prediction is correct.
TFLite (TensorFlow Lite)	A compressed, quantised model format optimised for inference on mobile and edge devices. Reduces model size by 2–4x with minimal accuracy loss.
Class Imbalance	Situation where training dataset has significantly more samples of one class than another. Addressed via class-weighted loss functions or oversampling.
Overfitting	Model memorises training data but fails to generalise to new unseen images. Detected by large gap between training and validation accuracy.

14.3 Stakeholder Sign-Off

Name	Role	Signature	Date
	Project Lead		
	AI/ML Engineer		
	Frontend Developer		
	Veterinary Consultant		

	QA Lead		

END OF DOCUMENT

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